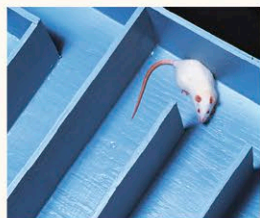


INTELLIGENCE

Intelligence itself is not easy to define, but indications of intelligence include the ability to learn, matched with behavioral flexibility and self-awareness, as well as the relative size of the brain to body size. Rats, for example, are considered to be highly intelligent because they can learn to perform new tasks, an ability that may be important when they first colonize a fresh habitat. Primates are considered to be the most intelligent of mammals. In some mammals, food-gathering appears to be related to intelligence: a deer feeding on plants has a brain relatively smaller than a cat that must “think” to outwit its prey.

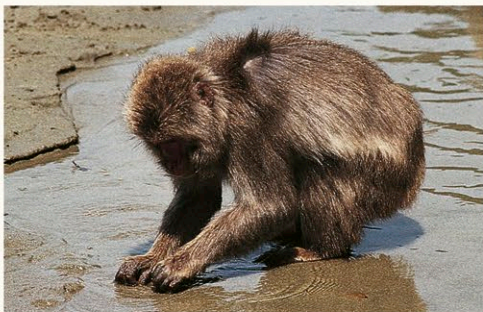


RAT IN A MAZE

Animal intelligence is often assessed by a creature's ability to complete simple tasks, usually using a food incentive. This albino brown rat is finding its way through a maze.

FOOD FOR THOUGHT

Primates are rare among mammals in that some can use a variety of objects as tools, while others clean their food prior to eating it. This Japanese macaque has learned to separate wheat grains from sand by immersing the food in water. The sand sinks; the wheat grains float on the surface.



in the groin) or in their urine or feces (which contain sexual hormones). They also communicate by body posture or facial gesture, by touch, and by sound, which may develop into complex messages.

Socialization begins at birth, when signals between a mother and her infant facilitate suckling. The process continues throughout the juvenile period, when the young interact with each other through play (see panel, opposite below) and gradually learn to interpret the behavior of adults.

Some mammals only interact to mate or to nurture young, but many form temporary or permanent social groupings beyond these minimal needs. The hoofed mammals,

for example, form herds for safety, because predators are more likely to be spotted. Interwoven with social behavior is the use of space: most mammals have a home range (the area within which an animal, or group, performs most of its activities), and in some these areas are developed into territories (an area that the animal, or group, defends against others of the same species).

Solitary mammals, such as aardvarks and most cats, usually occupy a territory that they defend against members of the same sex. A tigress, for example, will not permit other females to enter her territory, whereas the territory of a male tiger overlaps those of several females.

Some mammals, such as gibbons, live in monogamous pairs (with their immature offspring), and each pair occupies and defends a territory. However, recent studies indicate that the assumption of “lifelong fidelity” is wrong:

surreptitious copulations outside the pair are frequent, and many pairs break up and find new mates after some years.

Other mammals form larger social groups, of varied composition. In some species, such as seals and elephants, the sexes live separately for most of the year. Related females form social units, and the males either live alone or form small bachelor groups. In mammals living under this system there is ferocious competition among the males for the right to mate. The most

successful males are usually the largest and strongest individuals that also have the best weapons (antlers, horns, or tusks). In elephant seals and sea lions, as the females become sexually receptive, the males bellow and display to establish dominance.

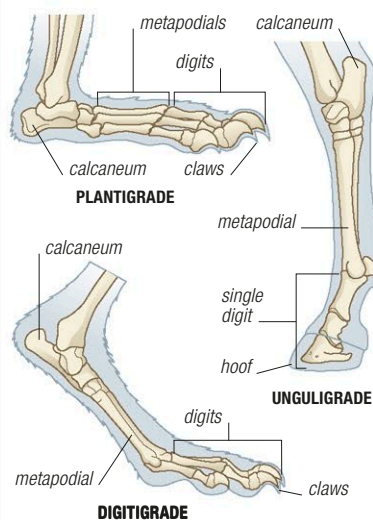
Other species, such as the plains zebra, form small, permanent harem groups, in which one male leads and mates with a group of females. The subordinate males are relegated to bachelor bands unless one can successfully challenge and expel a harem male.

The final type of larger social group exhibited by mammals is one in which the collection of individuals consists of several males and females. Such societies are restricted to primates and the social carnivores (although some other species, such as migrating wildebeests, form temporary mixed-sex groups). Among baboons, for example, several adult males join a troop of females related to each other (but not to the males) and compete for dominance and the right to mate with the females. These power struggles result in a continual change in the dominance hierarchy. In lion prides, the males are related to each other and cooperate in the defense of the females rather than competing for them. In wolf packs, only the dominant pair breeds; the other adults are the offspring from previous years who, instead of leaving to form their own packs, stay with their parents and help rear their younger siblings.

Locomotion

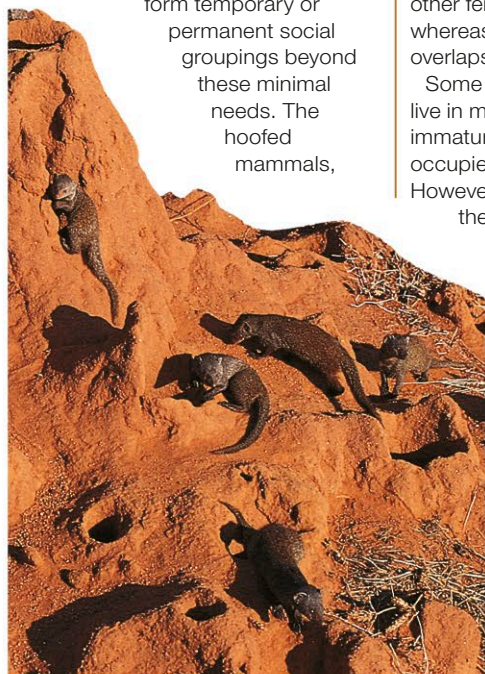
Mammals dwell in a huge variety of habitats and have accordingly developed a number of different ways of moving around. While a quadrupedal (4-footed) gait is the most common form of mammal locomotion, some species, such as kangaroos, have a bipedal (2-footed) gait. As mammals move, the way in which the

foot comes into contact with the ground varies in one of 3 different ways (see below). Locomotion is also related to the lifestyle of the animal. Predators, which require short, explosive bursts of speed to catch their prey, have a flexible spine. For example, in the cheetah—the fastest terrestrial animal—the backbone coils and uncoils with every stride, propelling the animal forward. It has been estimated that these movements increase the cheetah's speed by about 19mph (30kph). In contrast, prey species, such as gazelles, rely on speed and endurance to outrun predators. They have a rigid back, and the “energy cost” of their movement is reduced as a result of a lengthening of the lower segments of the limbs and a concentration of the muscles close to the body.



WALKING GAITS

When moving, plantigrade mammals, such as bears, keep the calcaneum (heel bone), metapodials, and digits (toes) of each foot on the ground. Digitigrade mammals, such as dogs, move with only the digits touching the ground. Unguligrade mammals, such as horses, walk and run on the tip of each digit.



BENEFITS OF COMMUNAL LIVING

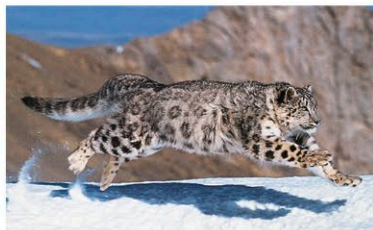
Living in a group has several advantages for small mammals, such as these dwarf mongooses. For example, there is more chance of spotting predators, the burden of rearing young can be shared, and the territory can be defended collectively.



AERIAL MAMMAL: LONG-EARED BAT



ARBOREAL MAMMAL: BLACK LEMUR



TERRESTRIAL MAMMAL: SNOW LEOPARD



AQUATIC MAMMAL: DOLPHIN

ADAPTABLE MAMMALS

Over time, the limbs of some mammals have evolved into wings, flippers, or grasping hands and feet, enabling bats to take to the air, whales and dolphins to adopt an aquatic existence, and primates to move freely through the trees. Predatory land mammals, such as cats, have developed a flexible spine for speed and maneuverability, and a long tail for balance.